Independent measurement at each isogeny-volcano level

Date: 2026-10-13

Status: **complete exploratory observation.** The protocol and workload were frozen before the fresh run. The sealed analyzer accepted all 392 jobs and all 160,000 planted-key certificates.

This is an exploratory additive replication. It is not ledger evidence, a sealed measured bound, an isogeny-transfer result, an end-to-end ECDLP speedup, or a weak-curve verdict.

Requested question and evidence status

- **Each sampled level:** equal-curve-weighted native negation-only counts at all four nodes for `m=37` and `m=41`; **verified complete**.
- **Large conductor factors:** four conditional vertical-edge contrasts and one factor interaction per lattice; **verified complete**.
- **Symmetry separation:** lower-level versus crater negation is primary; native level versus crater signed-`tau` is secondary context; **verified complete at the operation-count level**.
- **Uncertainty:** 50,000-replicate hierarchical bootstrap with pointwise and simultaneous intervals; **verified complete**.
- **Weakness:** explicit maps, subgroup preservation, recovery, and fully priced end-to-end costs are **not tested by this study**.

Outcome

All six lower-level versus crater-negation point estimates were between `-0.008978` and `+0.016988` operation-count bits. Every pointwise and simultaneous 95% interval contained zero. The three `m=37` simultaneous 90% ratio intervals were inside the frozen `[0.97, 1.03]` equivalence band. None of the three `m=41` intervals was contained in that band because the intervals were wider, so the predeclared **global equivalence result is false**. That is an inconclusive precision outcome at `m=41`, not evidence of a level effect.

Every conditional conductor-factor edge and both factor interactions also had a pointwise 95% interval containing zero. The fresh data therefore resolve no large-prime-factor difference at the tested precision, but do not establish a general conductor-independence law.

The native crater signed-`tau` context arm used 2.478--2.496 fewer counted-work bits than the native `m=37` arms and 2.683--2.702 fewer bits than the native `m=41` arms. This is a same-class action comparison, not a transported action, fully priced speedup, or weak-representative result.

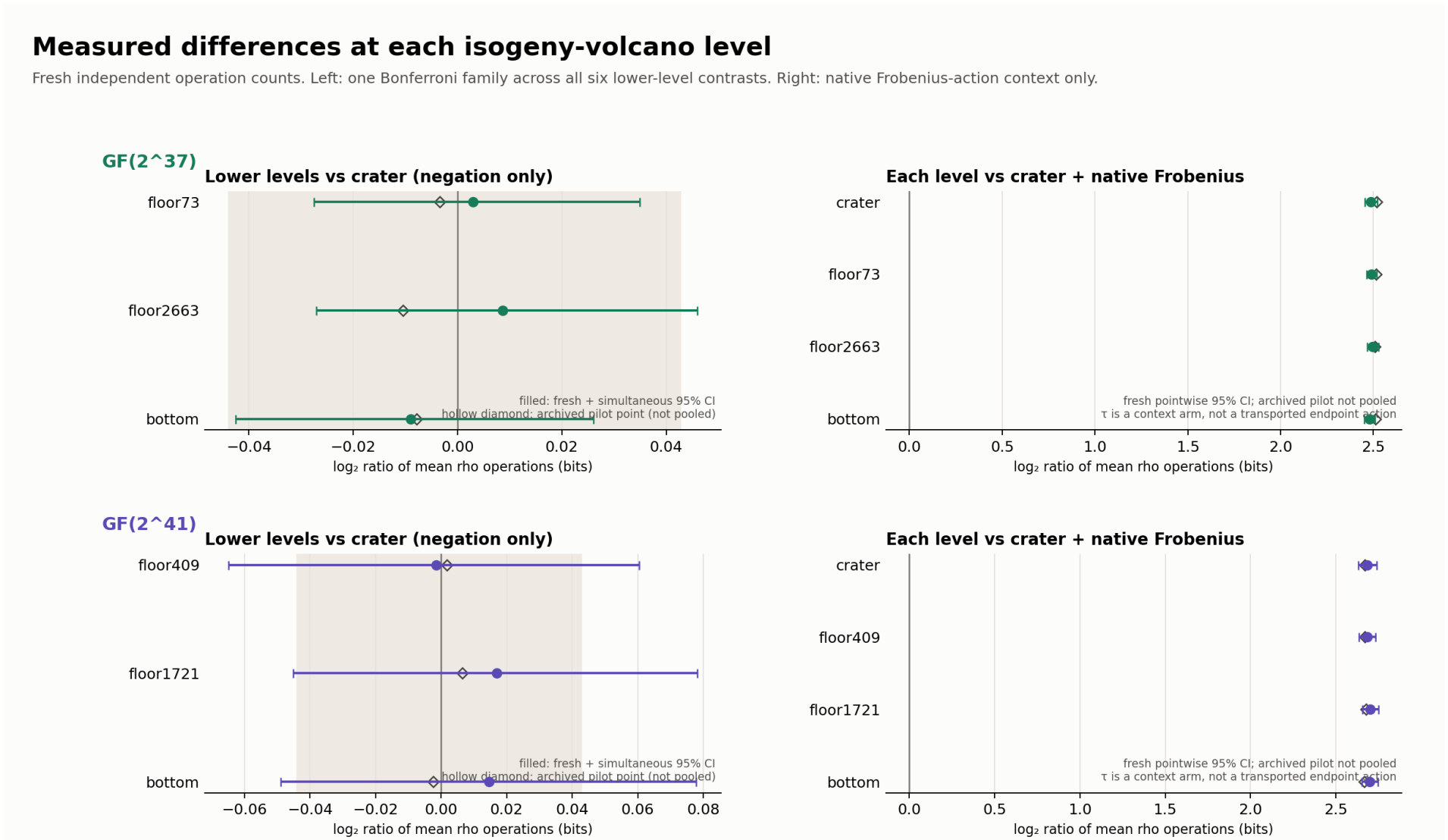


Figure source: plot.py; vector rendering: volcano-level-differences.svg. The left panels show simultaneous 95% error bars for the six-member Bonferroni family against the shaded +/-3% band. The frozen equivalence decision instead uses the simultaneous 90% intervals reported in the primary table below. The right panels are native Frobenius-action context only.

Objects and conductor gaps

The sampled nodes form conductor-factor lattices, not linear volcanoes. Write `Z[pi] = O_(f_pi) subseteq End(E) = O_(f_E) subseteq O_K`. The final column is the remaining index `[End(E):Z[pi]] = f_pi/f_E`; neither `f_E` nor this index is an ECDLP-bit discount.

study	node	`f_E`	`f_pi/f_E`	sampled curves	fresh negation-only solves
`m37`	crater	1	194,399	1	5,000
`m37`	floor73	73	2,663	74	74,000
`m37`	floor2663	2,663	73	22	22,000
`m37`	bottom	194,399	1	30	30,000
`m41`	crater	1	703,889	1	3,000
`m41`	floor409	409	1,721	30	6,000
`m41`	floor1721	1,721	409	30	6,000
`m41`	bottom	703,889	1	30	6,000

The crater also has a separate signed-Frobenius context arm: 5,000 fresh solves at `m=37` and 3,000 at `m=41`. Total frozen workload: 160,000 solves.

Fresh results

Displayed ratios, bit effects, and interval endpoints are rounded to six decimal places from `results.json` and `contrasts.csv`; those machine-readable files retain the analyzer's full precision.

Same-action level comparison: native negation only

This is the primary comparison. A ratio above one means that the named lower node used more counted group operations than the crater under the same native negation-only action. The six contrasts are one multiplicity family.

study	lower node / crater-neg	ratio of mean ops	`log2` ratio (bits)	simultaneous 95% CI (bits)	simultaneous 90% ratio CI	inside frozen +/-3% band?
`m37`	floor73	1.002024	+0.002917	[-0.027519, +0.034966]	[0.982683, 1.022510]	yes
`m37`	floor2663	1.005998	+0.008627	[-0.027028, +0.045977]	[0.983446, 1.029838]	yes
`m37`	bottom	0.993796	-0.008978	[-0.042598, +0.026042]	[0.973006, 1.015798]	yes
`m41`	floor409	0.999009	-0.001430	[-0.064901, +0.060353]	[0.959649, 1.039025]	no
`m41`	floor1721	1.011845	+0.016988	[-0.045113, +0.078181]	[0.972599, 1.051627]	no
`m41`	bottom	1.010235	+0.014691	[-0.048873, +0.077866]	[0.970339, 1.051116]	no

Global simultaneous equivalence within `[0.97, 1.03]`: **false**. All three `m37` contrasts pass the frozen rule; all three `m41` contrasts fail it on interval width. All six simultaneous 95% intervals include zero.

Removing the exact common target-dependent setup term changes none of the pointwise conclusions:

study	lower node / crater-neg	ex-setup ratio	ex-setup effect bits (pointwise 95% CI)
`m37`	floor73	1.002046	+0.002949 [-0.020710, +0.026881]
`m37`	floor2663	1.006064	+0.008722 [-0.018646, +0.036193]
`m37`	bottom	0.993728	-0.009077 [-0.034852, +0.016882]
`m41`	floor409	0.999009	-0.001430 [-0.049474, +0.044967]
`m41`	floor1721	1.011847	+0.016992 [-0.030108, +0.062563]

study	lower node / crater-neg	ex-setup ratio	ex-setup effect bits (pointwise 95% CI)
m41	bottom	1.010237	+0.014694 [-0.033223, +0.062030]

Per-level distributions and diagnostics

Means are equal-curve-weighted. Quantiles are across solves. Local-cap restarts, cycles, useless collisions, and descriptive timing remain in `level-summary.csv` rather than being collapsed into a single claim.

study	arm	curves	verified solves	mean ops	CV	median	q05--q95	mean / sqrt(r)	between-curve SD
m37	crater-neg	1	5,000	14,169.384	0.514915	13,438.5	3,720.75--27,507.15	0.933079	0*
m37	floor73	74	74,000	14,198.065	0.515446	13,357.0	3,797.00--27,468.00	0.934968	210.769
m37	floor2663	22	22,000	14,254.370	0.517279	13,358.5	3,779.00--27,707.00	0.938675	265.823
m37	bottom	30	30,000	14,081.484	0.515192	13,259.5	3,740.00--27,188.05	0.927290	232.380
m37	crater-tau	1	5,000	2,526.806	0.477009	2,393.5	818.95--4,683.60	0.166395	0*
m41	crater-neg	1	3,000	679,711.961	0.520689	643,740.0	174,979.85--1,314,890.30	0.916726	0*
m41	floor409	30	6,000	679,038.566	0.520734	638,614.5	179,390.35--1,319,431.15	0.915818	26,628.454
m41	floor1721	30	6,000	687,762.852	0.523548	645,044.0	174,911.30--1,333,054.75	0.927585	25,148.441
m41	bottom	30	6,000	686,668.633	0.529098	645,431.0	174,657.45--1,353,669.10	0.926109	27,556.208
m41	crater-tau	1	3,000	105,717.736	0.524878	99,427.5	25,925.20--206,454.85	0.142581	0*

* The between-curve SD is zero by construction for a one-curve arm.

The remaining frozen diagnostics are below. Timing is `L0` descriptive only; it is not a performance bound. Cycle, local-restart, and useless-collision rates are per 1,000 walk starts.

study	arm	mean walk starts	cycles / 1k	local restarts / 1k	useless / 1k	mean sec / solve	aggregate ns / counted op
m37	crater-neg	421.601800	0.601990	0	0.000949	0.007142646	504.090
m37	floor73	422.360216	0.593640	0.000032	0.001056	0.007207290	507.625
m37	floor2663	423.894727	0.580333	0	0.001287	0.007336559	514.688
m37	bottom	418.818133	0.605434	0	0.001433	0.007129208	506.282
m37	crater-tau	71.452400	0.316294	0	0	0.002517758	996.420
m41	crater-neg	20,433.160000	0.532925	0	0.038565	0.375741313	552.795
m41	floor409	20,414.360167	0.538714	0	0.038282	0.377169809	555.447
m41	floor1721	20,675.288333	0.545506	0	0.040378	0.376377932	547.250
m41	bottom	20,643.024500	0.536452	0	0.039360	0.375495268	546.836
m41	crater-tau	3,177.196333	0.003252	0	0.007449	0.121365455	1,148.014

Does the large prime conductor factor differ?

The direct measurements are the conditional vertical edges. Each factor is added once with the other factor absent and once with it present. The interaction is `log2(mu_bottom) - log2(mu_factor1) - log2(mu_factor2) + log2(mu_crater)`.

study	factor	exact comparison	ratio	effect (bits)	pointwise 95% CI (bits)
m37	73	floor73 / crater	1.002024	+0.002917	[-0.020485, +0.026585]
m37	73	bottom / floor2663	0.987871	-0.017605	[-0.038453, +0.003022]

study	factor	exact comparison	ratio	effect (bits)	pointwise 95% CI (bits)
m37	2,663	floor2663 / crater	1.005998	+0.008627	[-0.018443, +0.035801]
m37	2,663	bottom / floor73	0.991789	-0.011895	[-0.027377, +0.003650]
m37	73 x 2,663	bottom x crater / (floor73 x floor2663)	0.985876	-0.020522	[-0.052015, +0.010702]
m41	409	floor409 / crater	0.999009	-0.001430	[-0.049462, +0.044956]
m41	409	bottom / floor1721	0.998409	-0.002297	[-0.040039, +0.036423]
m41	1,721	floor1721 / crater	1.011845	+0.016988	[-0.030100, +0.062547]
m41	1,721	bottom / floor409	1.011237	+0.016121	[-0.022858, +0.055663]
m41	409 x 1,721	bottom x crater / (floor409 x floor1721)	0.999399	-0.000867	[-0.060528, +0.061079]

Every interval in this table contains zero. Thus no factor edge or interaction was resolved as nonzero at pointwise 95% confidence. These contrasts answer only whether native negation-only counts differ at the tested lattice edges. Two lattices, non-random curve samples, and independent targets do not identify a causal conductor-size law, establish monotonicity, or support extrapolation to the conductor factors of a named NIST curve.

Symmetry/action context: native level versus crater signed-`tau`

This secondary contrast intentionally answers a different question. It mixes the ring-level change with loss of the crater's cheap native Frobenius action, so it must not be called a conductor effect. Both total operations and the predeclared setup-excluded sensitivity are reported.

study	native arm / crater-tau	total ratio	total effect bits (pointwise 95% CI)	ex-setup ratio	ex-setup bits (pointwise 95% CI)
m37	crater-neg	5.607627	2.487391 [2.454008, 2.520821]	5.908741	2.562851 [2.528131, 2.597772]
m37	floor73	5.618978	2.490308 [2.463909, 2.517348]	5.920833	2.565800 [2.537818, 2.594553]
m37	floor2663	5.641261	2.496018 [2.465855, 2.526040]	5.944572	2.571573 [2.539883, 2.603116]
m37	bottom	5.572840	2.478413 [2.449994, 2.506916]	5.871680	2.553773 [2.523878, 2.583864]
m41	crater-neg	6.429498	2.684706 [2.631132, 2.740854]	6.438088	2.686632 [2.633020, 2.742836]
m41	floor409	6.423128	2.683276 [2.636225, 2.731547]	6.431709	2.685202 [2.638105, 2.733533]
m41	floor1721	6.505653	2.701694 [2.655006, 2.748727]	6.514363	2.703624 [2.656874, 2.750706]
m41	bottom	6.495302	2.699397 [2.652320, 2.747287]	6.503997	2.701327 [2.654196, 2.749277]

The setup-excluded sensitivity subtracts exactly 155 operations at `m=37` and 167 at `m=41`. The `ops` counter does not price field operations used by orbit canonicalisation, curve/generator precomputation, final verification, or any isogeny map. These rows are therefore operation-count context, not a fully priced speedup or transfer result.

Archived pilot context (not pooled)

The historical point estimates below motivated the fresh replication. They are observations from different seed namespaces and archived runs, not fresh outcomes and not inputs to the new bootstrap.

study	lower node	archived lower/crater-neg ratio	archived effect (bits)	fresh minus archived (bits)
m37	floor73	0.997670	-0.003365	+0.006283
m37	floor2663	0.992785	-0.010446	+0.019074
m37	bottom	0.994579	-0.007843	-0.001135
m41	floor409	1.001240	+0.001788	-0.003218
m41	floor1721	1.004482	+0.006451	+0.010536
m41	bottom	0.998343	-0.002392	+0.017083

Archived sources:

- `research/isogeny_volcano_rho_20261005/results.json`, SHA-256 `2eb7caee726587c6d5450c81afee5275f2d39c96edef3514443e05d145e2f452`;
- `research/isogeny_volcano_rho_m41_20261006/results.json`, SHA-256 `11c53edde3011a0ef81cbab0e7a62390cb20d1be07ed83814bba4f4219dc9235`.

Statistical method and limitations

The primary estimand is the ratio of equal-curve-weighted mean `ops`, not the mean of per-solve ratios. The frozen analysis uses 50,000 fixed-seed three-stage empirical bootstrap replicates: curves, process chunks within selected curves, and solves within selected chunks. The one-curve crater arms resample chunks and then solves. The implementation uses exactly 4,096 fixed-seed solve-resampled means per chunk as an empirical reservoir, with a 2,000,000-cell memory cap.

Pointwise 95% intervals describe individual contrasts. Bonferroni simultaneous 95% intervals cover the frozen family of six primary level contrasts. Bonferroni simultaneous 90% intervals implement the frozen +/-3% equivalence rule. Equivalence inside that margin is not equality, and a confidence interval containing one is not evidence of equivalence.

The curve samples were fixed historically, but targets are independent rather than paired across curves because the legacy solver shares one PRNG stream for curve setup, target generation, and walk randomness. This design can test reproducibility and quantify node-level differences in this implementation; it cannot causally attribute a difference to conductor alone.

Weakness claim boundary

This study can support only statements about measured source-operation counts on these proxy classes. It does not provide an explicit isogeny map or dual, subgroup-preservation certificate, endpoint action, scalar recovery path, map-evaluation cost, or complete cold-start/amortised cost. Consequently:

- a conductor gap alone is not evidence of ECDLP weakness;
- a lower operation count at one sampled node would be a lead requiring replication and complete pricing, not a weak-curve assertion;
- a native crater signed-`tau` advantage does not establish that the action is cheaply transported to a lower representative;
- a bounded null result supports only **no weakness found in scope**, never absence of a weak representative or transfer in general.

The labels **weak representative exists**, **source-transfer-weak**, or **class-weak** remain unavailable unless their separate map, subgroup, recovery, and total-cost obligations are discharged.

Evidence and provenance

All repository paths and commit identifiers cited in this report have provenance `internal`; no recalled or unverified external source supports a result claim here.

Frozen at commit `f10e03e3d8b0e584d23202deb391c5e7b06026cf`. `FROZEN.sha256` seals the protocol, specification, runner, analyzer, and plot source. The producer worktree was clean at that commit before both binaries were built with GCC 14.2.0 using `-O2 -march=native`. Binary SHA-256 values were `15ae904dc8f3313b1902c57bb6acd2e7e19d12f824148f2ee74e58ca` for `m37` and `a2c80bdc011d8e8c157b46d91c2dd9a2b768f242bfa239da49d26c1457467c76` for `m41`.

The run began at `2026-10-13T06:10:25.597160+00:00` and completed at `2026-10-13T06:53:06.059611+00:00` on an AMD EPYC 9V74 host, using four workers pinned to CPUs 0--3. The runner recorded 392/392 completed jobs, 160,000/160,000 verified solves, and no error. The analyzer verified the exact file sets and receipts, found zero duplicate rows, and accepted every planted certificate. The `m37` raw tree contains 272 files with SHA-256 `bae5bbe0f77da5c376a43f18a9641e27bc3d95488e8f1a86bb8c62cac64075f8`; the `m41` tree contains 120 files with SHA-256 `0096c753b7e96a5977318482c87455ee9d3166c381b46dcad9644e5f73b54226`.

artifact	SHA-256
`run1.tar.zst`	`37d8abf2d12d51eed53ef013f05bbf8a17028ee51c60355cf1e2f2994fa63eb7`
`results.json`	`cc52d06c8e04769ceed6e194abbef651f0c6cee330670452c5e98027451ea372`
`level-summary.csv`	`61370aff72167634f9f3c0e9e667183ecef5abc18530ad8b831f6bef96e34757`
`contrasts.csv`	`09e1a2b20e24efecee2906b7058f4447a5ea063992310eb662fce0c989c01681`
`volcano-level-differences.svg`	`6a2790432ca1f8ee30f249723e4d5031eedf94c4d02d250be58ac45339699812`
`volcano-level-differences.png`	`0d677b680140d6197f68f26a7a52f144236f1a6fd51dfa9fae180dec3d4a3f71`

The plan, receipt, and session hashes recorded inside `results.json` are, respectively, `d54036bdcc762f8c56434755be7a786a7756fdf5ff75b9406bb77b0cb88de2f1`, `9bbd38db211cdd496f1fc0993` and `76947b5b4e310768534783d465dde37e3ea2cc891443dd7bcd96b70db3d9ad8e`.

ATTEMPTS.md preserves the launch and archive history. A clean-room extraction was byte-equal to the live run tree. Analysis of a relocated extraction was intentionally rejected because absolute output paths are part of the frozen receipts; reproducible reanalysis must recreate the recorded run-root path. This is an archive portability limitation, not a failed planted-key certificate.

Primary frozen sources are PROTOCOL.md, spec.json, run_replication.py, analyze.py, and plot.py.

Reproduce

The frozen analyzer is path-bound. From the repository root, recreate the recorded run root if it is not already present, then analyze and plot:

```
study_dir=research/isogeny_volcano_level_replication_20261013
run_root=/tmp/isogeny-volcano-level-replication-20261013-run1
if [ ! -d "$run_root" ]; then
  mkdir "$run_root"
  tar --zstd -xf "$study_dir/run1.tar.zst" -C "$run_root"
fi
python3 research/isogeny_volcano_level_replication_20261013/analyze.py \
  --run-root "$run_root" \
  --output-dir "$study_dir"
MPLCONFIGDIR=/tmp/matplotlib-volcano-level \
  python3 research/isogeny_volcano_level_replication_20261013/plot.py
```

Build a byte-reproducible PDF with the installed Pandoc 3.1.11.1 and pdfTeX toolchain. The trailer ID is derived from the report source rather than from wall-clock state:

```
report_source=research/isogeny_volcano_level_replication_20261013/README.md
report_sha256=$(sha256sum "$report_source" | awk '{print $1}')
report_id=${report_sha256:0:32}
SOURCE_DATE_EPOCH=1791849600 \
FORCE_SOURCE_DATE=1 \
pandoc "$report_source" \
  --from=gfm --standalone --pdf-engine=pdflatex \
  --resource-path=research/isogeny_volcano_level_replication_20261013 \
  -V documentclass=extarticle -V fontsize=9pt \
  -V geometry:landscape -V geometry:margin=0.55in -V colorlinks=true \
  -V "header-includes=\\pdftrailerid{<${report_id}><${report_id}>}" \
  -o research/isogeny_volcano_level_replication_20261013/report.pdf
```

Validate the rendered file and extract its text for inspection:

```
pdfinfo research/isogeny_volcano_level_replication_20261013/report.pdf
pdftotext -layout \
  research/isogeny_volcano_level_replication_20261013/report.pdf -
```